

NIGAM MEHTA

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Education

Shah & Anchor Kutchhi Engineering College, Mumbai
B.Tech, Electronics Engineering – VLSI Design and Technology

Aug 2023 – 2028

Projects

8-bit Pipelined Processor

GitHub

- Built an 8-bit 3-stage pipelined processor (Fetch/Decode/Execute) in Verilog with an 8-register file, ALU (ADD, ADDI, SUB, SLL, HALT), and hardware hazard detection/stall logic.
- Synthesized and implemented on FPGA (Xilinx Vivado/XSim) at 100MHz with full timing closure, using only 95 LUTs and 104 registers (<2% utilization).

Modular Digital Data Monitoring Unit (SoC IP Core)

GitHub

- Designed an 8-bit RTL monitoring unit with a Moore FSM (Idle, Alarm, Cooldown), sticky alarm, fault capture register, and dual-mode signed/unsigned comparison logic.
- Scaled into an AMBA APB4 slave IP with a 5-register memory-mapped interface; built AutoArchitect in Python to automate Yosys synthesis and netlist SVG generation.

Rail Nova – Railway Reservation Simulator

GitHub

- Built a FastAPI + SQLAlchemy/SQLite reservation system modeling 8,990 stations and 5,277 trains, replicating the CNF/RAC/Waitlist allocation algorithm with auto-promotion on cancellation.
- Added real-time PNR/train tracking via WebSockets, JWT auth, and an in-app e-Wallet.

Bank Management System

GitHub

- Built a full-stack banking simulator: C++ backend (unordered_map for O(1) lookups, custom linked lists for transaction history) served via a Flask REST API to a web frontend.
- Implemented account creation, deposits/withdrawals, loans, UPI integration, and file-based persistence across sessions.

Professional Experience

Microchip Embedded Systems Developer Intern, AICTE-EduSkills (with Microchip)

Jun – Aug 2026

Completed an 8-week PIC16 microcontroller track: MPLAB X, embedded C (callbacks, modular design), BLE app development, Azure IoT sensor nodes with motor control, and real-time data visualization.

VLSI Design & Semiconductor Engineering Intern, EduSkills Foundation

Apr – Jun 2026

Studied digital IC architectures, RTL design methodologies, ASIC verification flows, and EDA tooling.

Data Analyst Intern, Elevate Labs

Feb – Mar 2026

Cleaned/analyzed 10,000+ records (Python, Pandas, Power BI), built 5 dashboards, and automated reporting – cutting manual effort by 30%.

Android Developer Intern, EduSkills Foundation

Oct – Dec 2025

Developed 3 Android app modules (Java/Kotlin) covering UI screens and core app logic.

AI & Machine Learning Intern, EduSkills Foundation / Edunet Foundation

Jun – Sep 2025

Trained and evaluated ML classification models (scikit-learn) on labeled datasets, reaching 88% accuracy across preprocessing, training, and evaluation stages.

Technical Skills

VLSI/Hardware: Verilog, RTL Design, FSM, ASIC Verification, AMBA APB4, FPGA Synthesis (Xilinx Vivado), Yosys, Digital Logic Design

Embedded: PIC16 MCUs, MPLAB X, Embedded C, BLE/IoT, Azure IoT

Software/AI: Python, C++, SQL, Flask, FastAPI, SQLAlchemy, Scikit-learn, SHAP, SMOTE, Pandas, Power BI, Git/GitHub, Android (Java/Kotlin)

Certifications

Junior Cybersecurity Analyst Career Path (Cisco) | Generative AI in Action (IBM SkillsBuild) | Cybersecurity Fundamentals (IBM SkillsBuild) | Embedded System Design Overview (Maven Silicon) | Full list on portfolio (19 total)